Track 1 pipeline report

Team fulopjoz. MVA Hackathon 2026, Track 1, variant prioritization. Submission file: `submission_track1.csv`.

Result

The pipeline ranks the clinically confirmed genotype first of 8,666 candidates: compound-heterozygous **BUB1B p.Leu737Ter** (chr15:40,209,701 T>G, stop gained; ClinVar allele ID 529272) plus **BUB1B p.Asn1002Lys** (chr15:40,220,612 T>G, missense), score 0.5823. Score at rank 2 is 0.5243. The score column `epcr` — estimated probability of causal relationship, the challenge-defined (0, 1] score — equals the pipeline ranking score.

Rank	Genotype	Score
1	BUB1B p.Leu737Ter + p.Asn1002Lys	0.5823
2-7	RAI1 frameshift-deletion pairs (two orientations)	0.5243 / 0.4914
8-10	KIF7 pairs (founder allele is a common stop polymorphism, gnomAD frequency 0.561)	0.4515
2,373 and below	homozygous-recessive singles (ordered after biallelic pairs by design)	—

Method

One pass over the 5,012,204-record proband VCF. Each stage's counter is measured from the run.

1. **Quality filter.** FILTER is PASS or absent, genotype quality at least 20, read depth at least 10, alternate-allele read fraction at least 0.25. Keeps 4,592,140.
2. **Effect calling.** Variants are annotated against local GENCODE v47 MANE transcripts. Stop gained, frameshift, and splice-site changes count as loss of function (LoF); missense and in-frame indels are scored separately. Keeps 16,963 coding or splice candidates.
3. **Rarity.** Allele frequency from local gnomAD v4.1 exome sites; every record at a multi-record position is merged and alleles match exactly on (chrom, position, ref, alt). ClinVar joins as background evidence only.
4. **Candidate pairing.** Compound-heterozygous pairs (an LoF allele plus a second damaging allele in the same gene; phase not required), rare homozygous missense, rare homozygous stop gain, rare homozygous in-frame indels. Finds 2,372 pairs and 6,294 homozygous candidates.
5. **Scoring.** Each variant scores as a weighted mix: founder effect 0.45, predicted pathogenicity 0.60, rarity 0.30, plus phenotype similarity to the eight case HPO terms, weight 2.0. A pair scores as the stronger allele × (0.5 + 0.5 × the weaker).

External cross-check

Exomiser 14.0.0 on the same VCF ranks BUB1B first (p = 0.0001) with the same eight HPO terms; without phenotype input its top is a 46-way tie that still contains BUB1B. The records of both runs (`exomiser_summary.tsv` on the analysis host, artifact hashes verified 2026-09-17) are traceable; the answer is therefore not an artifact of this pipeline's own code.

The phenotype weight's influence is characterised the same way: at weight 0 the true pair ranks 25 of 8,666 (epcr 0.6913); at the shipped weight 2.0 it ranks 1 (record: `desensitization.json` on the analysis host, verified 2026-09-18). We disclose this dependence rather than hide it.

Reproduction

Python 3.12 with pysam, cyvcf2, pyyaml. References (paths in `pipeline/conf/track1_config.yml`): GENCODE v47 MANE, GRCh38 FASTA, gnomAD v4.1 exome sites, ClinVar, HPO with the OMIM gene-to-phenotype map. Pathogenicity comes from consequence class; CADD v1.7 scores are used when local tiles cover a site, and none did in this run, so the CADD term is neutral and all CADD fields in the ranked table are empty.

```
python3 pipeline/scripts/prioritize.py \
    --conf pipeline/conf/track1_config.yml \
    --with-af \
    --with-hpo \
    --out output/annot_full.tsv
python3 pipeline/scripts/make_submission_csv.py \
    --in output/annot_full.tsv \
    --out submission_track1.csv
```

The annotate and HPO modules carry built-in self-tests (`python3 pipeline/scripts/annotate.py`, `python3 pipeline/scripts/_hpo.py`); they print SKIP with the missing reference named when a host lacks the reference files.

Disclosure

Built with AI assistance (Kimi K3 via Oh My Pi); no patient data left the analysis hosts and none was sent to any external API, per the challenge rules.

Data availability

Four artefacts cited above and in the technical documentation are **not present in this repository, by design**. They are pipeline outputs computed from the patient's sequence data, so each one is a per-variant table for a single identifiable individual. Publishing them would disclose case data that this project's whole architecture exists to contain, and the repository becomes public after the hackathon closes.

Cited artefact	Why it is absent
`desensitization.json`	Per-variant ranks for the case under varied phenotype weighting. Patient-derived.
`exomiser_summary.tsv`	Per-variant Exomiser output for the case. Patient-derived.
`output/annot_full.tsv`	The full annotated variant table for the case. Patient-derived.
`annot_ns.tsv`	The nonsynonymous subset of the same table. Patient-derived.

They were used to produce the numbers reported above and are retained on the whitelisted analysis machine. They can be made available to the organisers or to a reviewer under the data agreement that governs the source dataset, on request.

`hpo_gene_to_phenotype.tsv` is also referenced and is likewise not shipped, for a different and more ordinary reason: it is a public reference file distributed by the Human Phenotype Ontology project and is fetched rather than vendored.

Everything needed to re-run the pipeline is here: the scripts under `pipeline/`, the configuration under `pipeline/conf/`, and the reproduction steps above. What is withheld is the output computed from one child's genome, not the method that produced it.